

Assignment 1.1 — Project Synopsis

Project Title: Deep Generative Image Inpainting

1. Problem Statement & Motivation

Image inpainting is the task of reconstructing missing or corrupted regions of an image using the surrounding visual context. It has applications in image restoration, object removal, photograph editing, and recovery of damaged visual content. The major challenge is to generate realistic content that is also semantically consistent with the surrounding image, particularly when the missing region is large or has an irregular shape.

Traditional deep learning approaches, including CNN- and GAN-based inpainting methods, generally require task-specific training and can struggle to reconstruct meaningful structures for large or complex missing regions. Recent diffusion models have demonstrated strong generative capabilities and can provide a powerful prior for reconstructing missing visual content.

This project focuses on using a pretrained diffusion model to fill in masked/missing regions of an image without task-specific retraining, exploring how mask shape, mask size, and the number of resampling steps affect the realism and semantic consistency of the completed region. The primary methodology will be based on RePaint: Inpainting Using Denoising Diffusion Probabilistic Models (Lugmayr et al., CVPR 2022), which modifies the reverse diffusion sampling process of a pretrained diffusion model to incorporate known image regions. The diffusion-based approach will be benchmarked against a GAN-based inpainting baseline such as LaMa.

As an extension, the project will investigate whether the fixed resampling strategy can be improved through adaptive resampling, where the number of diffusion refinement steps is adjusted according to the difficulty of the masked region. Related approaches such as GradPaint, which also performs training-free diffusion inpainting through inference-time guidance, will be studied to position and refine the proposed approach.

2. Proposed Methodology & Pipeline

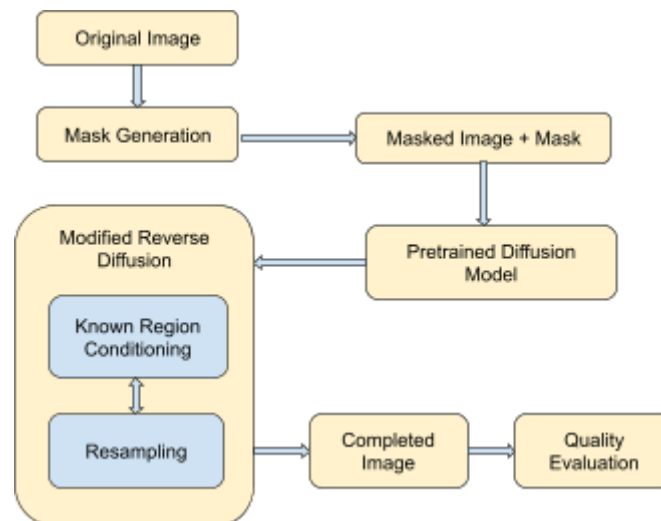


Fig 1: Inpainting System Architecture Block Diagram

As illustrated in Figure 1, the original image will first be resized and normalized, after which different masks with varying shapes and sizes will be generated to simulate missing regions. The masked image and corresponding mask will then be provided to a pretrained diffusion model without performing task-specific retraining. The reverse diffusion process will be modified following the RePaint methodology, where the known pixels are repeatedly incorporated during denoising while the missing region is generated by the diffusion model. Different numbers of resampling steps will be evaluated to study their effect on reconstruction quality.

The experiments will be conducted using datasets such as CelebA-HQ and Places2, allowing evaluation on both structured face images and diverse natural scenes. The generated images will be evaluated using PSNR, SSIM, LPIPS, and FID, along with inference time and the number of diffusion model evaluations. A GAN-based baseline such as LaMa will be used for comparison.

The study will further investigate an adaptive resampling strategy in which computational effort is dynamically allocated according to factors such as mask size, mask complexity, and reconstruction consistency. The objective is to determine whether adaptive sampling can achieve comparable or improved image quality with fewer diffusion steps.

Related works: RePaint (Lugmayr et al., 2022) provides the primary foundation for training-free diffusion inpainting; GradPaint explores gradient-guided training-free diffusion inpainting; BrushNet and PowerPaint investigate other forms of diffusion-based inpainting with additional conditioning or controllability; and LaMa provides a non-diffusion, trained inpainting baseline. These works will be used as references and comparison points rather than all being implemented.

3. Hyperparameters & Experimental Setup (subject to change during implementation)

Parameter	Planned Setup
Dataset	CelebA-HQ + Places2
Diffusion Model	Pretrained DDPM used by RePaint
Primary Method	RePaint-style training-free inpainting
Baseline	LaMa (GAN/non-diffusion inpainting baseline)
Mask Sizes	Small, medium, large, and extreme masks
Mask Shapes	Rectangular, square, irregular/free-form masks
Resampling Steps	Multiple settings to study quality vs. computation
Evaluation Metrics	PSNR, SSIM, LPIPS, FID
Efficiency Metrics	Inference time + Number of Function Evaluations (NFE)
Proposed Extension	Adaptive/difficulty-aware resampling
Hardware	GPUs Environments(Kaggle/Collab/Local GPUs)

4.Key References:

- 1)Lugmayr et al. — “RePaint: Inpainting Using Denoising Diffusion Probabilistic Models,” CVPR 2022. — Primary paper and methodology.
- 2)Suvorov et al. — “Resolution-Robust Large Mask Inpainting with Fourier Convolutions,” WACV 2022. — LaMa baseline.
- 3)GradPaint — Training-free diffusion-based inpainting using inference-time gradient guidance.
- 4)BrushNet — Diffusion-based inpainting using an additional masked-image conditioning branch.
- 5)PowerPaint — Controllable and versatile diffusion-based image inpainting.